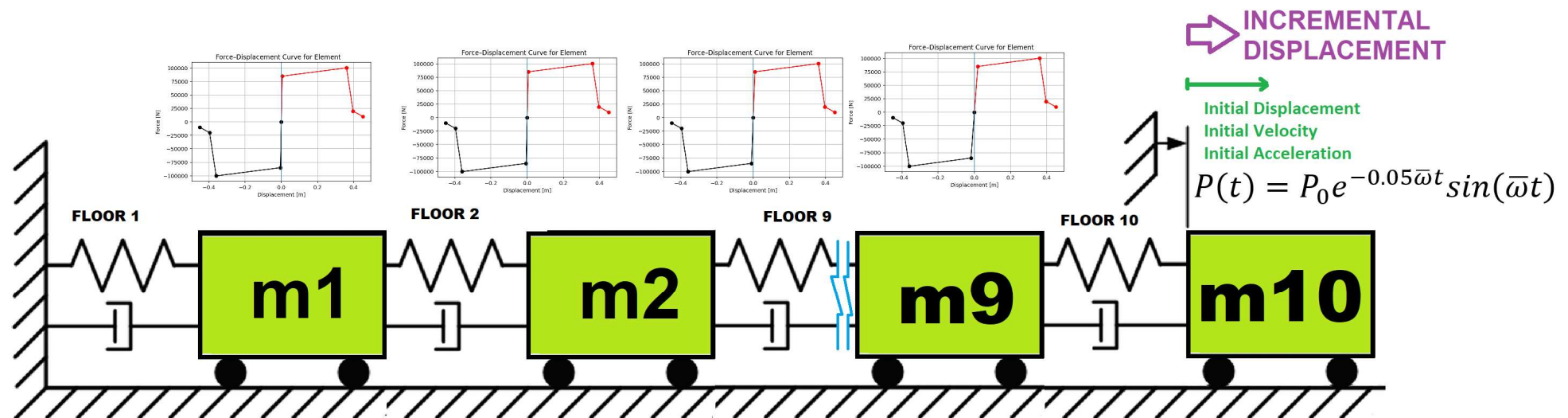


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

PUSHOVER ANALYSIS OF A MULTI-DEGREE-OF- FREEDOM STRUCTURE VIA EVALUATION OF A MULTILINEAR FITTING CURVE

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

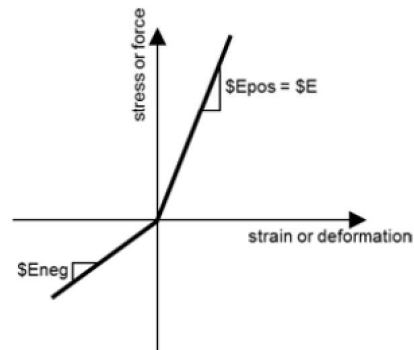
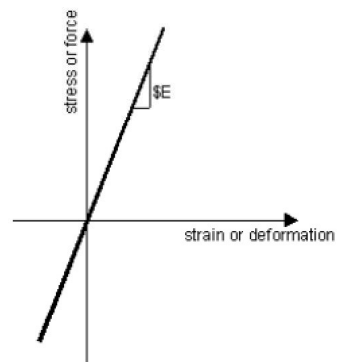
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

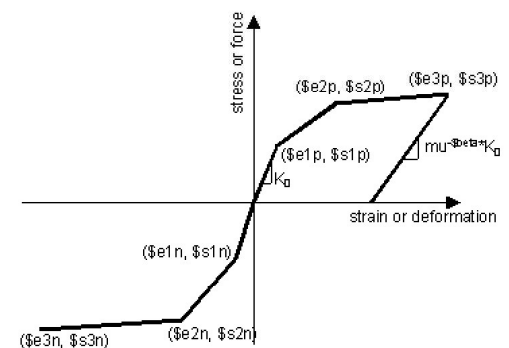
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

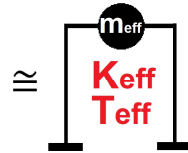
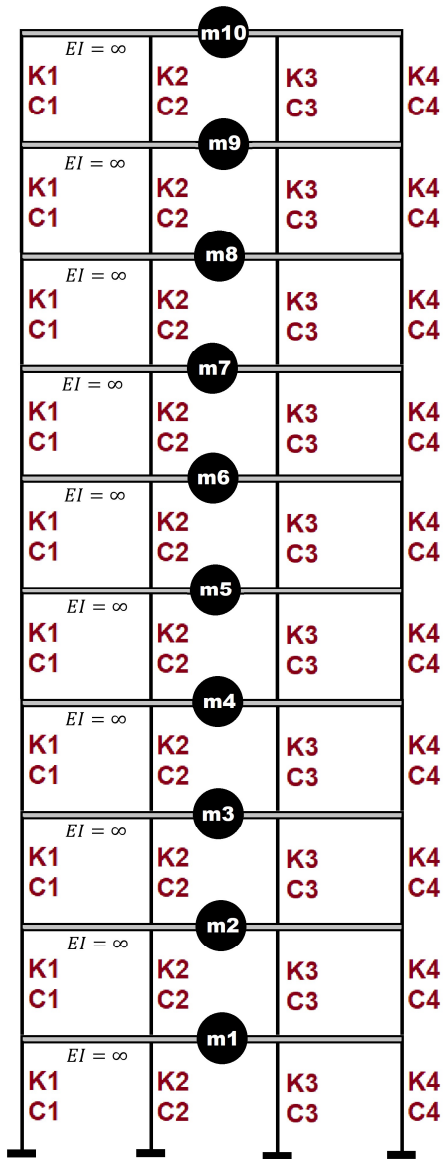
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



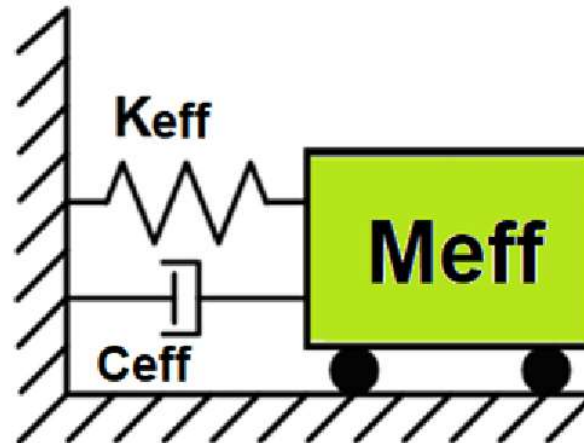
INELASTIC MATERIAL



Displacement-based seismic analysis transformation,
converting a multi-degree-of-freedom (MDOF) system
into an equivalent single-degree-of-freedom (SDOF)
system for seismic assessment

$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



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PUSHOVER_MDOF_10_FLOORS_04_COLUMNS.py

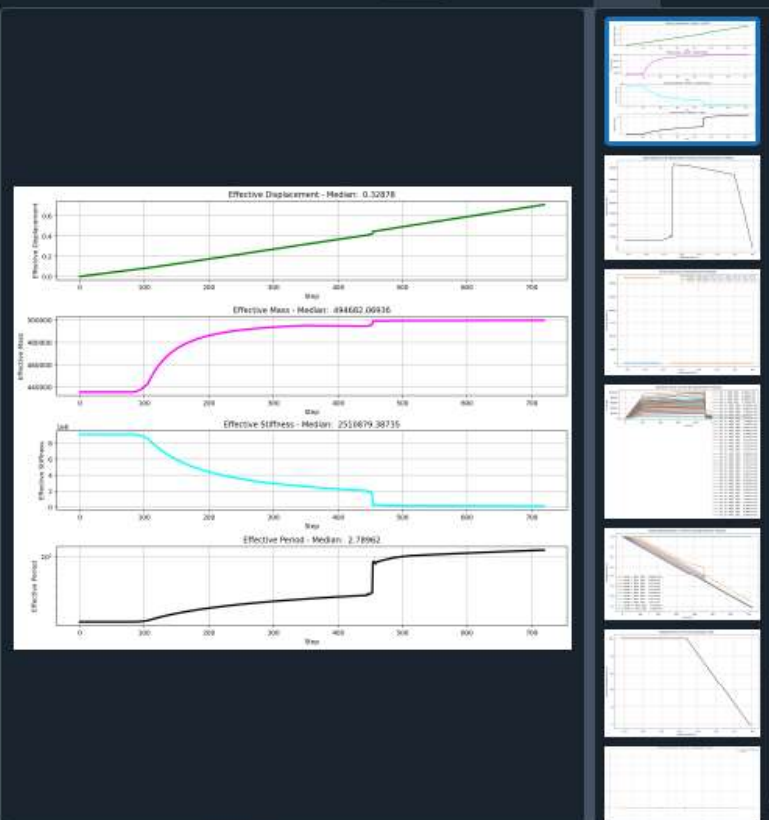
```

1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # PUSHOVER ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM STRUCTURE VIA EVALUATION OF A MULTILINEAR FITTING CURV
4 # -----
5 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANAL
6 # -----
7 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
8 # EMAIL: salar.d.ghashghaei@gmail.com
9 #####
10 """
11 This Python script performs nonlinear pushover analysis of a 10-story MDOF
12 building frame using OpenSeesPy to derive an equivalent SDOF system for
13 displacement-based seismic design.
14 1. It models a 10-DOF lumped-mass shear building with 4 parallel zeroLength
15 springs per floor, each combining a hysteretic uniaxial material (elastic or inelastic)
16 and a viscous damper.
17
18 2. Spring properties (yield force, ultimate force, elastic stiffness, ultimate displacement)
19 are defined per column, from which yield displacement, hardening ratio, and damping coefficient are com
20
21 3. Multiple analysis types are supported: STATIC, PUSHOVER, CYCLIC_DISPLACEMENT, FREE-VIBRATION,
22 SEISMIC, and external time-dependent loading (static or dynamic).
23
24 4. In the PUSHOVER case, displacement control is applied at the top node with small increments
25 up to twice the ultimate displacement, recording base reaction, nodal displacement, element
26 forces, and damage index at each step.
27
28 5. At every step, eigenvalue analysis (via `EIGENVALUE_ANALYSIS_FUN`) recomputes the structure's
29 period (min/max) to track stiffness degradation.
30
31 6. The damage index is computed using a ductility-based function (`DAMAGE_INDEX_FUN`)
32 comparing current displacement to yield and ultimate displacements.
33
34 7. Results are visualized: base shear vs. displacement, period evolution, element forces

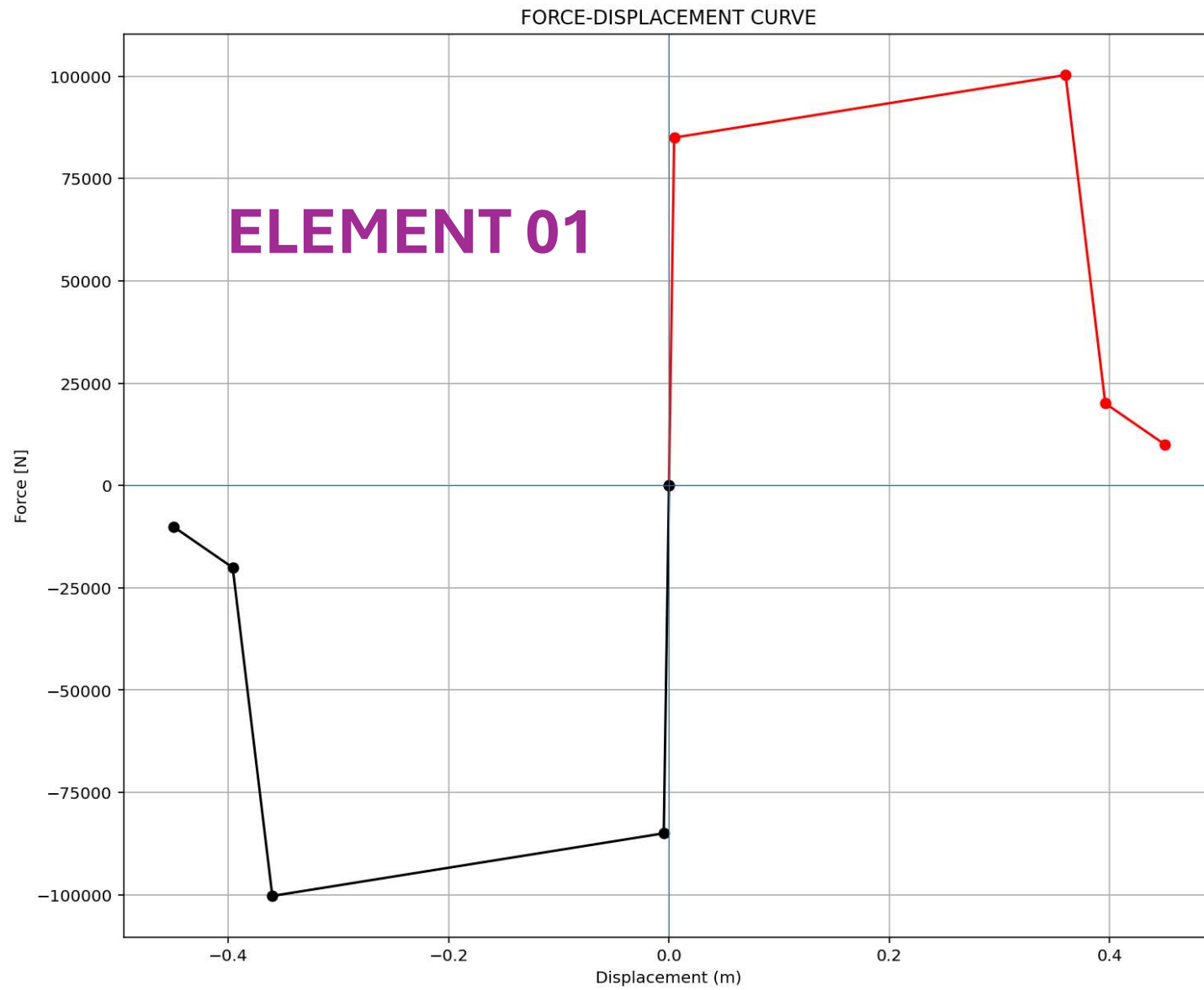
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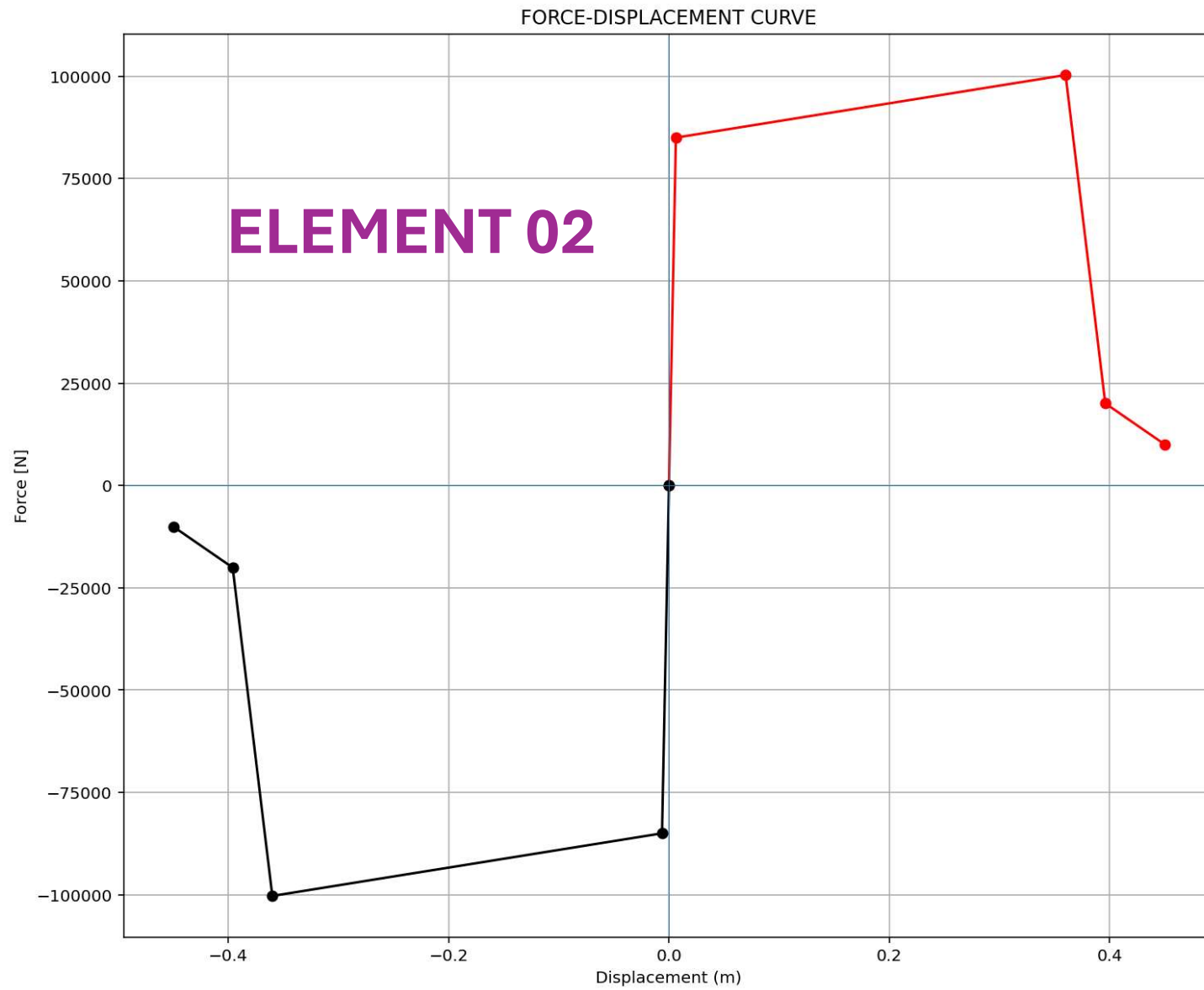
...ORS_04_COLUMNS_8_ANA\PUSHOVER_MDOF_10_FLOORS_04_COLUMNS

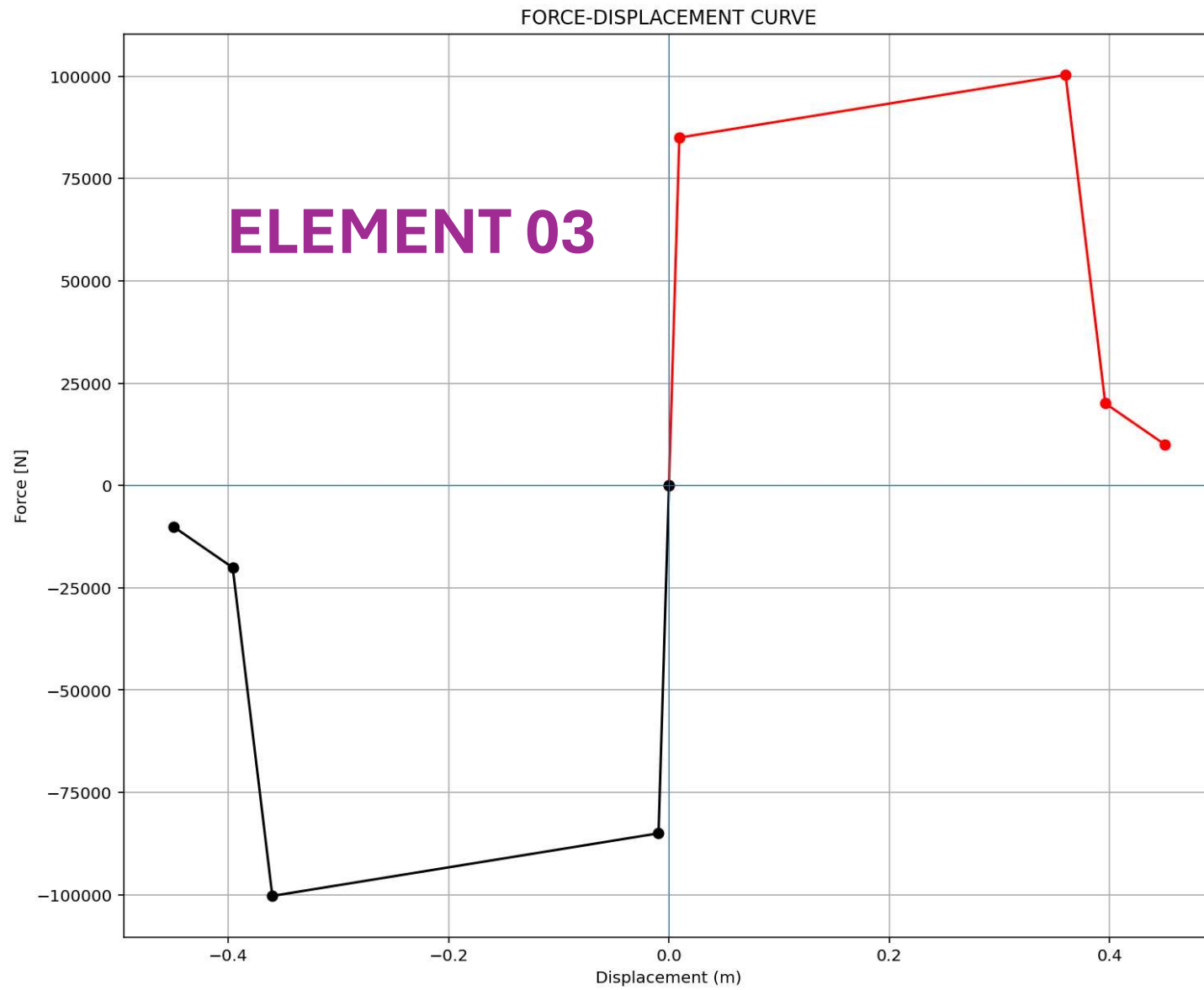
21 %

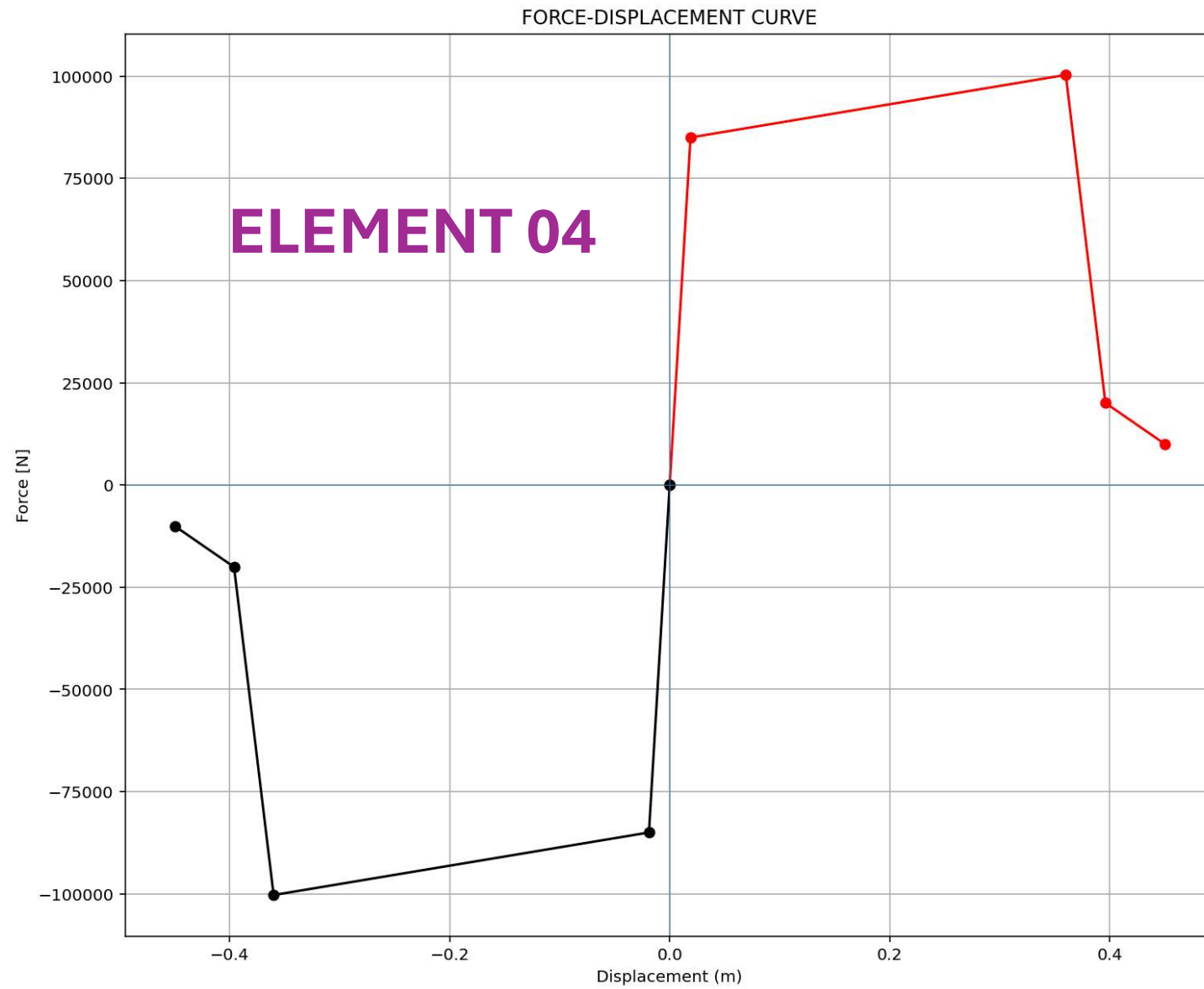


IPython Console Plots Variable Explorer Help Debugger History Files









PUSHOVER ANALYSIS

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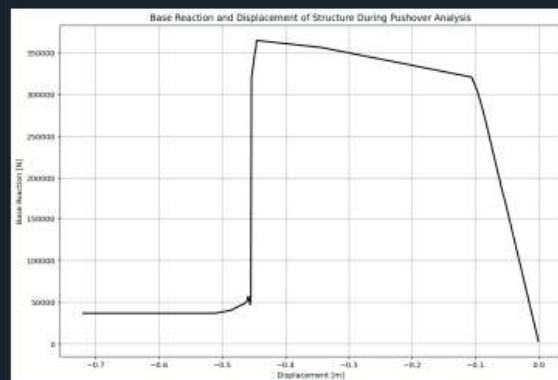
PUSHOVER_MDOF_10_FLOORS_04_COLUMNS.py

```

1 #####
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```

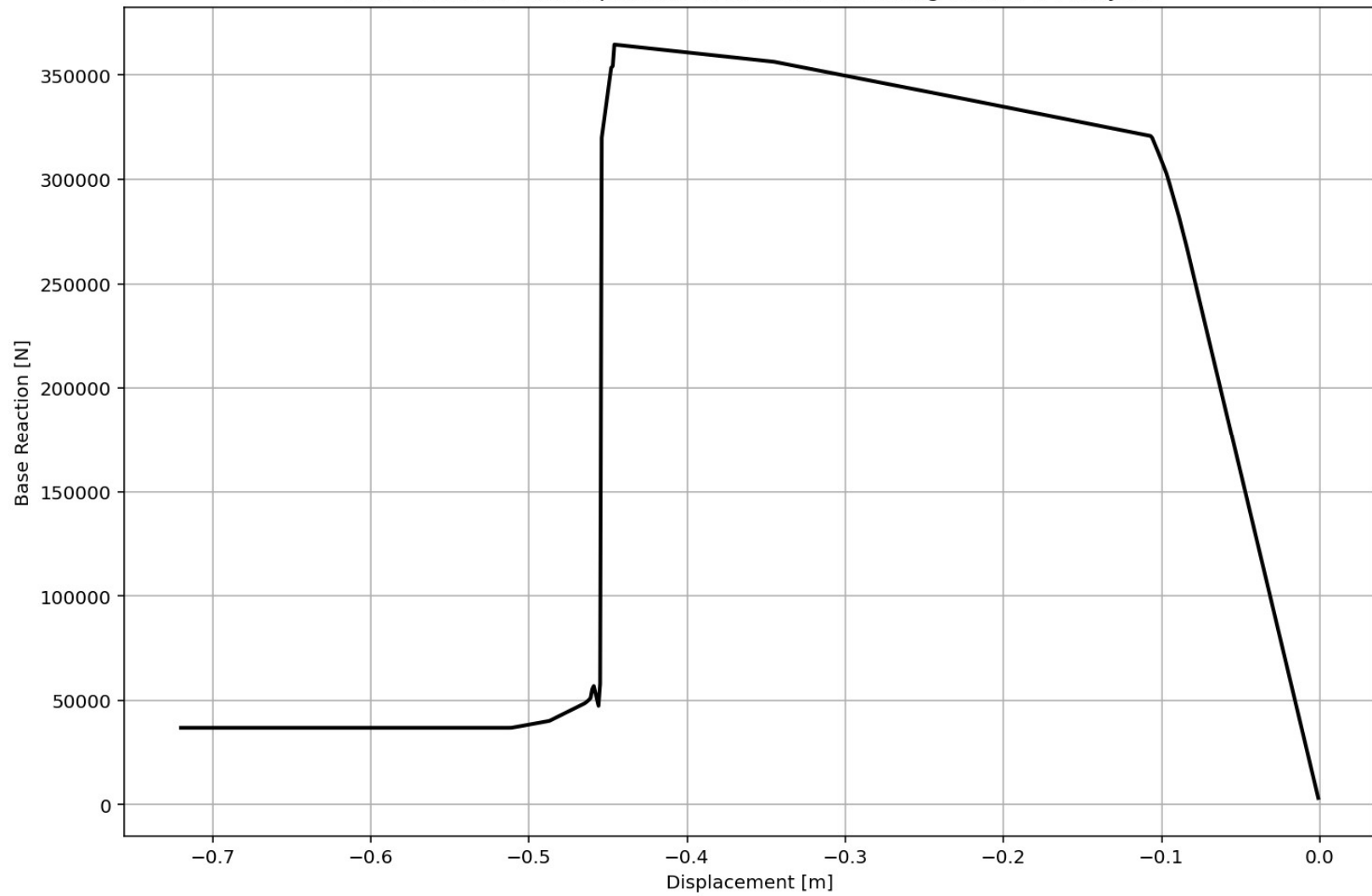
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24 %

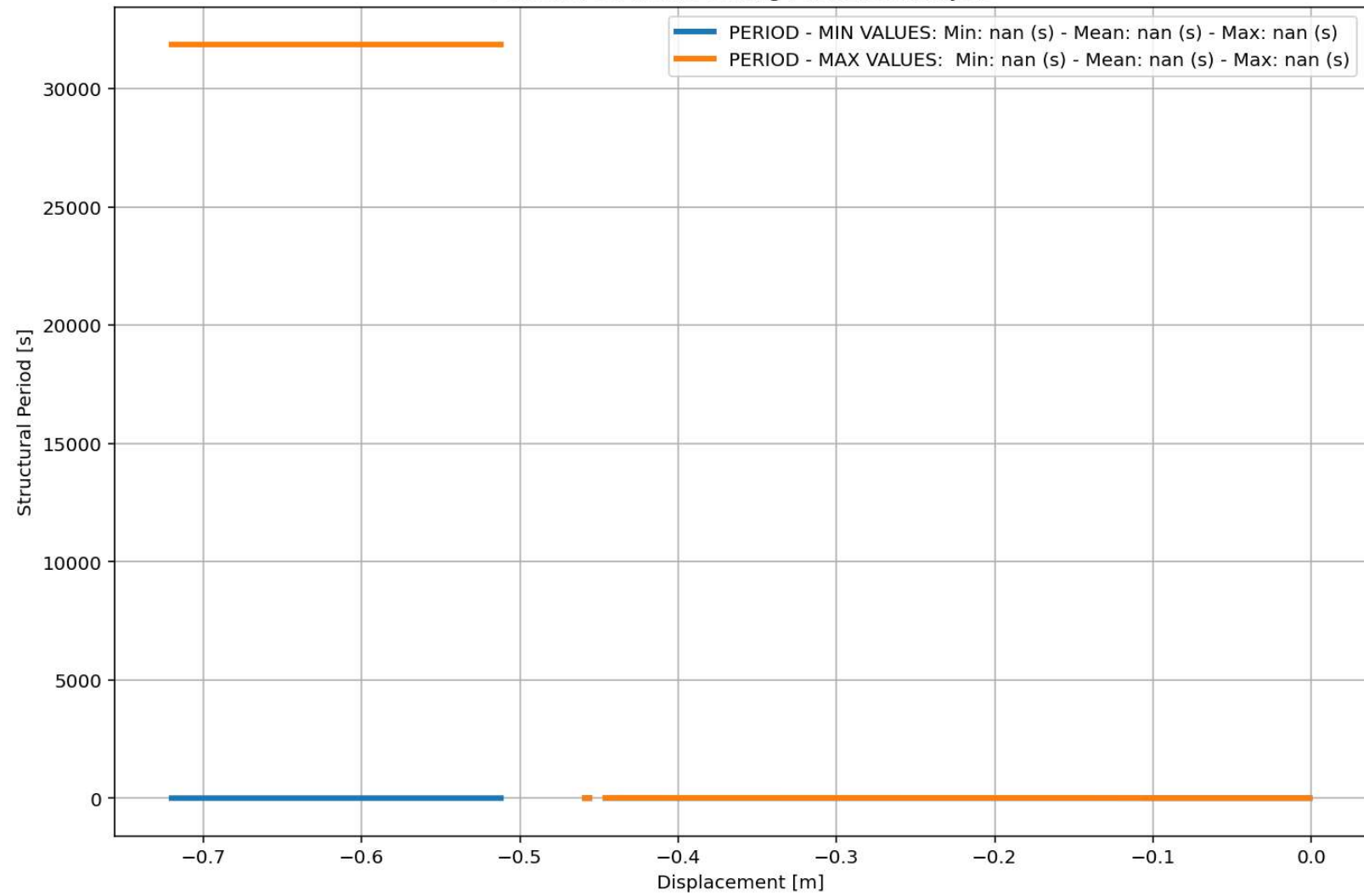


IPython Console Plots Variable Explorer Help Debugger History Files

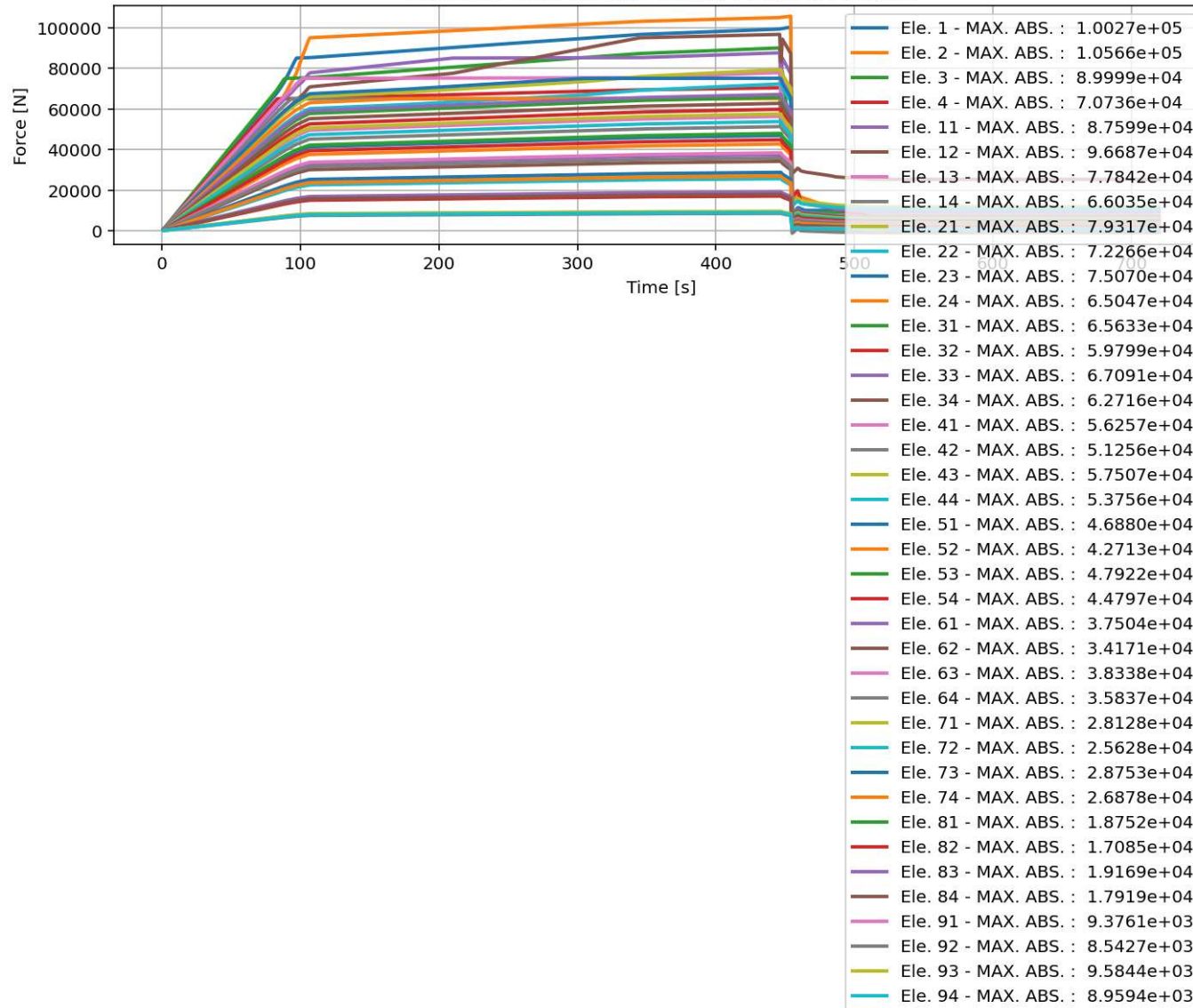
Base Reaction and Displacement of Structure During Pushover Analysis



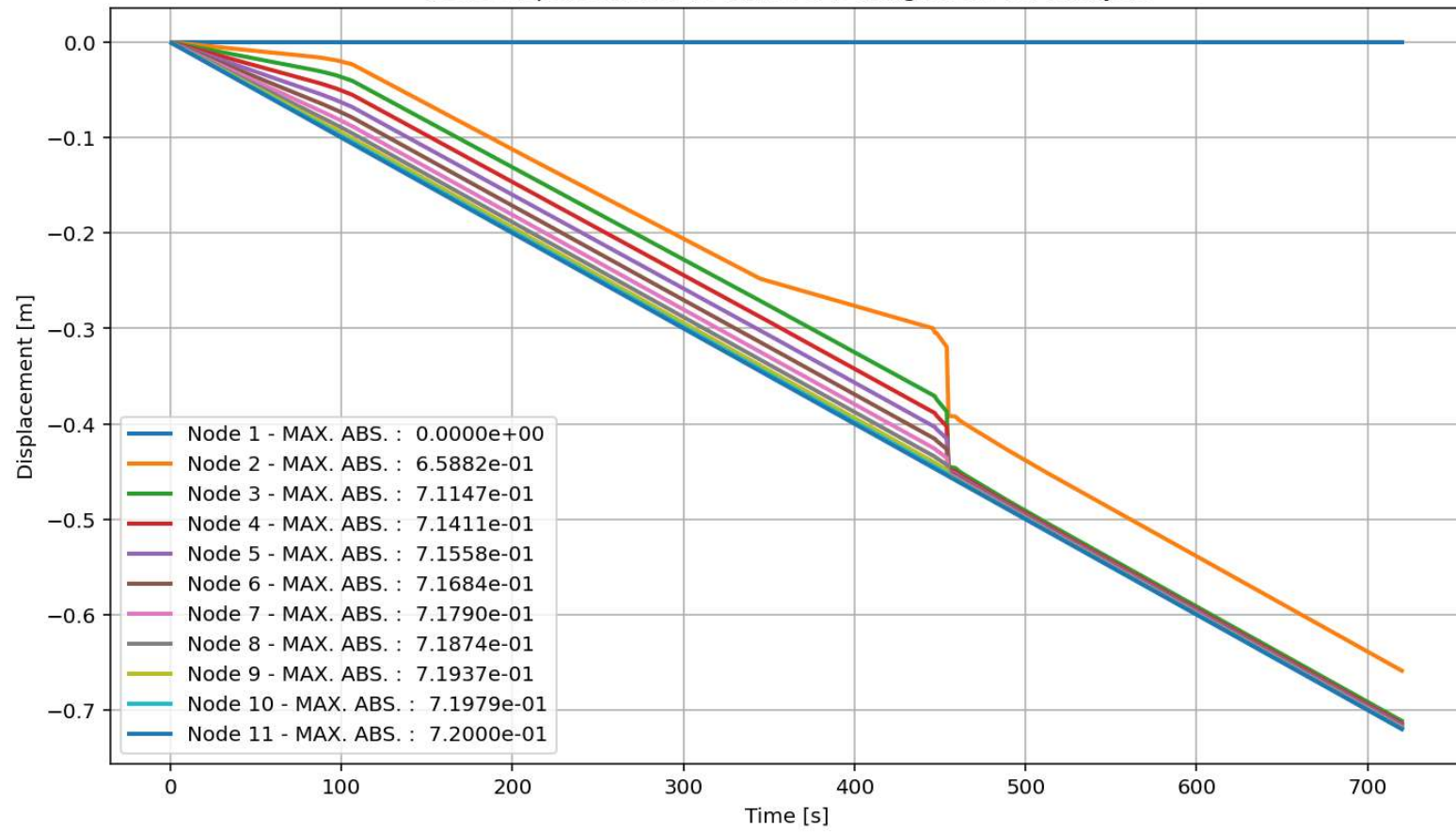
Period of Structure During Pushover Analysis

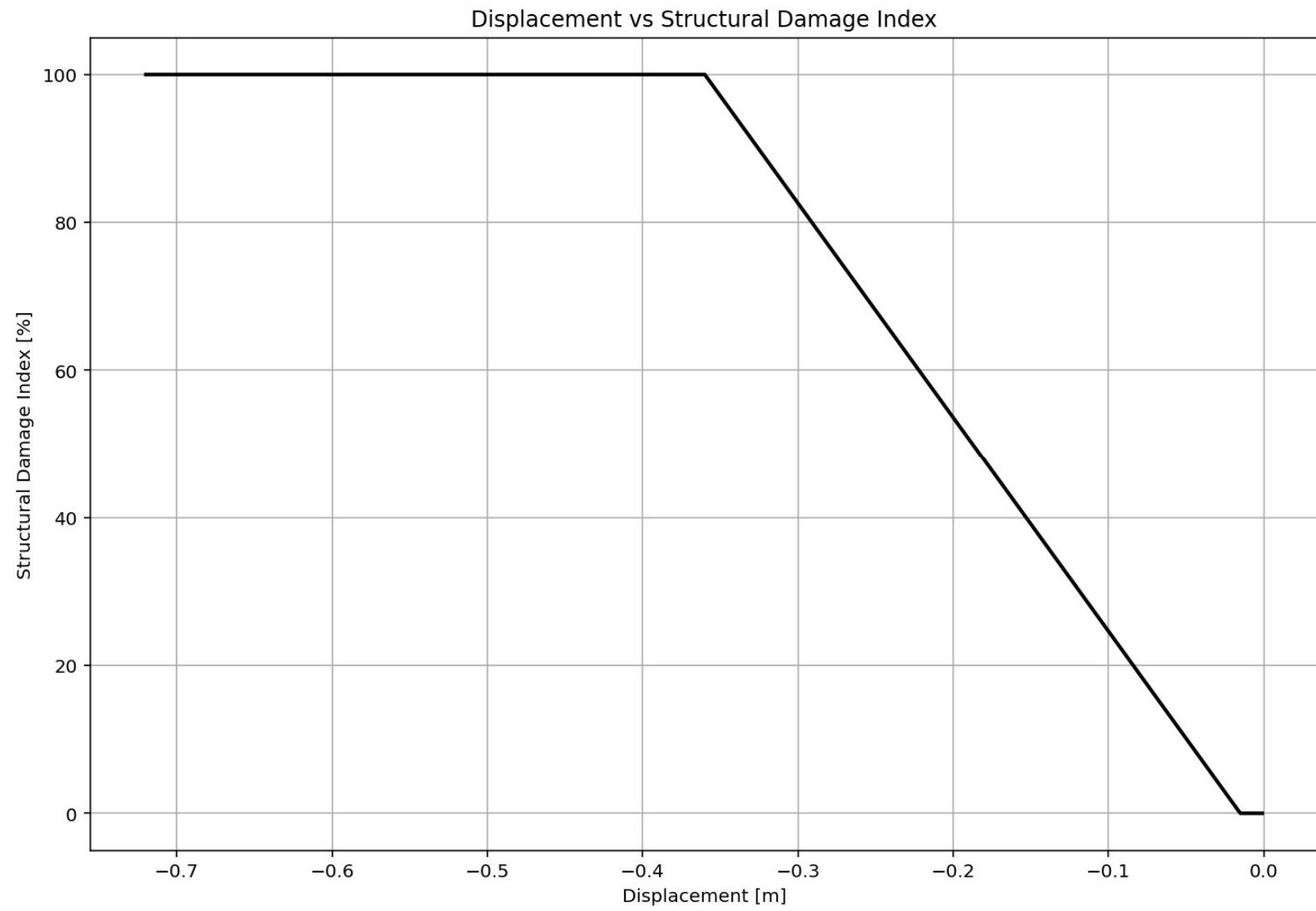


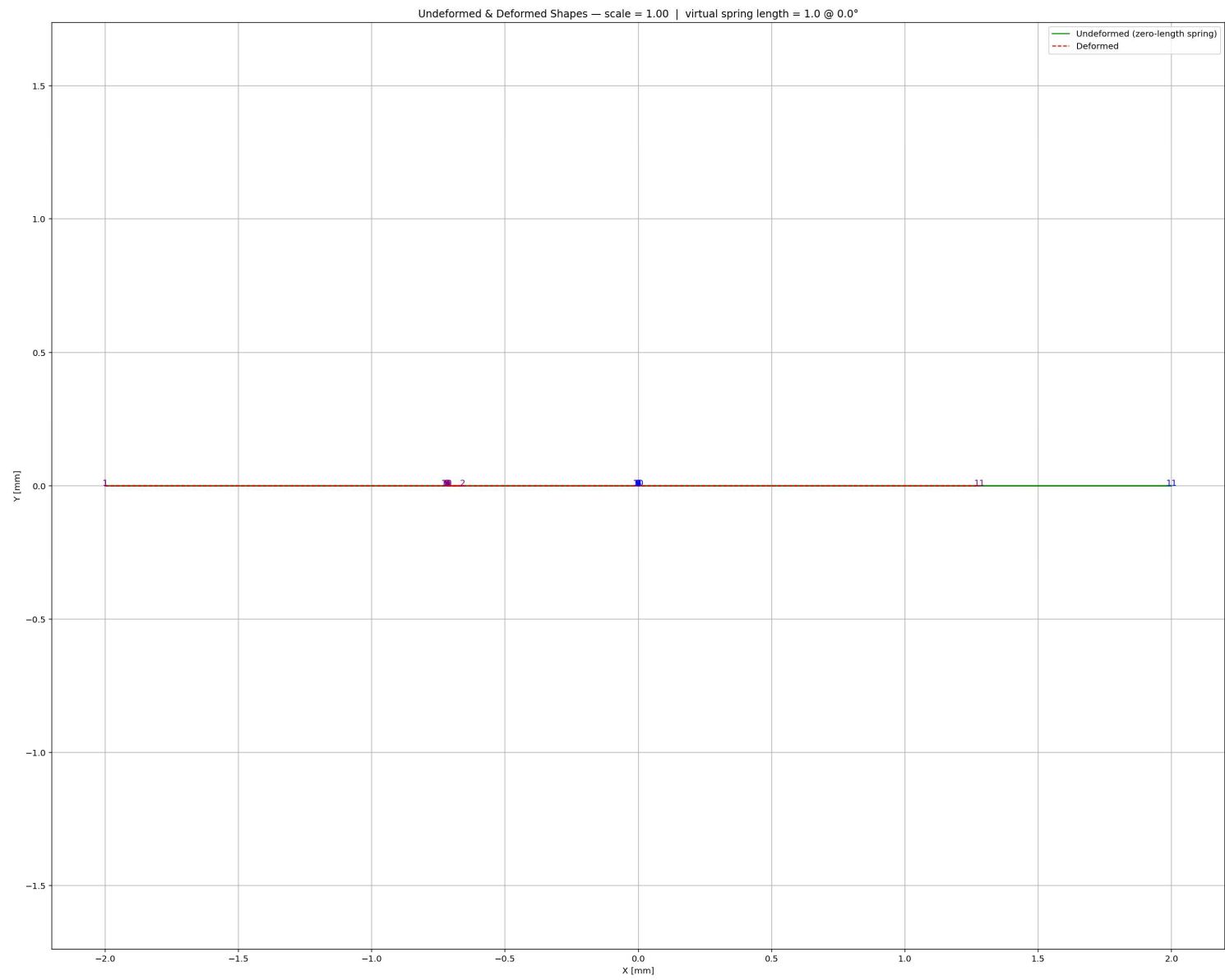
elements force vs Time During Pushover Analysis



Node Displacements vs Time for During Pushover Analysis





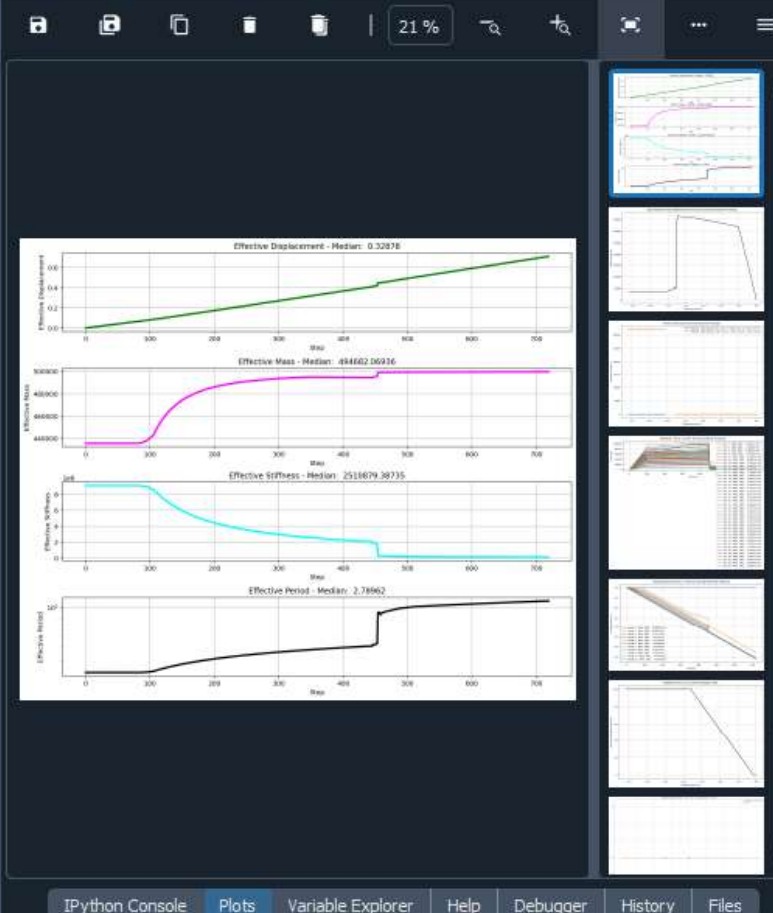


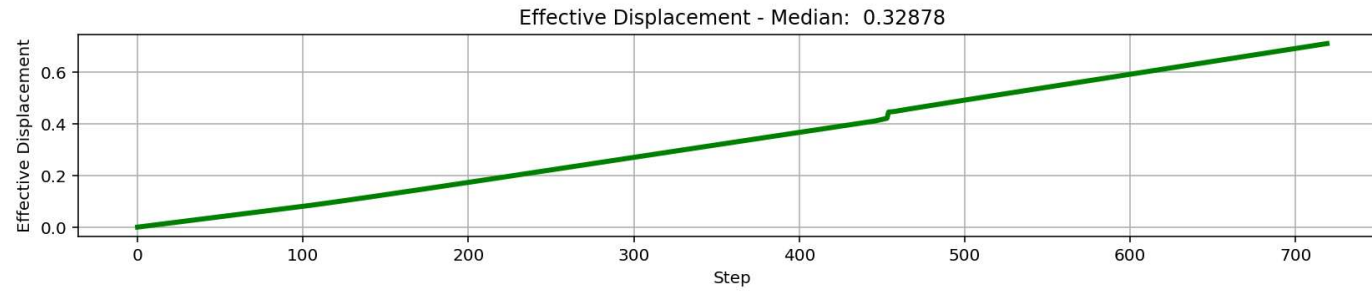
C:\Users\Dell\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...UMNS\COMPUTE_EFFECTIVE_PROPERTIES_FUN_PUSHOVER.py

```
PUSHOVER_MDOF_10_FLOORS_04_COLUMNS.py x COMPUTE_EFFECTIVE...ES_FUN_PUSHOVER.py x

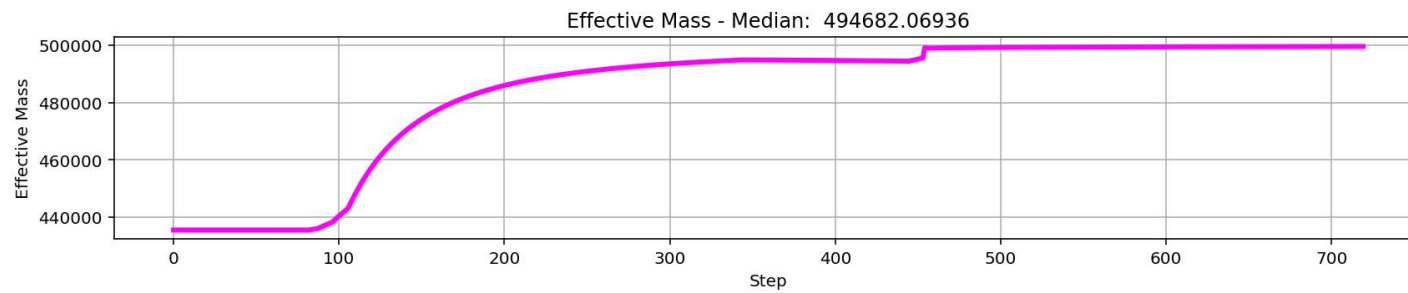
1  """
2  #%% EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANAL
3  # Change MDOF to SDOF System with Displacement Based Design Concept
4  # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
5  """
6  This script implements a displacement-based pushover transformation,
7  converting a multi-degree-of-freedom (MDOF) system into an equivalent
8  single-degree-of-freedom (SDOF) system for seismic assessment.
9  It calculates effective modal properties-displacement, mass, and
10 stiffness-by weighting element forces and nodal displacements according
11 to a presumed deformed shape.
12 The derived effective period provides a simplified dynamic characteristic
13 for performance-based engineering. The visualizations effectively track the
14 evolution of these equivalent parameters throughout the nonlinear analysis steps.
15 """
16 #-----
17 displacement_X_1 = np.array(list(node_displacements.values())[1]) # DOF 02
18 displacement_X_2 = np.array(list(node_displacements.values())[2]) # DOF 03
19 displacement_X_3 = np.array(list(node_displacements.values())[3]) # DOF 04
20 displacement_X_4 = np.array(list(node_displacements.values())[4]) # DOF 05
21 displacement_X_5 = np.array(list(node_displacements.values())[5]) # DOF 06
22 displacement_X_6 = np.array(list(node_displacements.values())[6]) # DOF 07
23 displacement_X_7 = np.array(list(node_displacements.values())[7]) # DOF 08
24 displacement_X_8 = np.array(list(node_displacements.values())[8]) # DOF 09
25 displacement_X_9 = np.array(list(node_displacements.values())[9]) # DOF 10
26 displacement_X_10 = np.array(list(node_displacements.values())[10]) # DOF 11
27
28 ele_force_01 = np.array(list(ele_force.values())[0]) # ELEMENT 01
29 ele_force_02 = np.array(list(ele_force.values())[1]) # ELEMENT 02
30 ele_force_03 = np.array(list(ele_force.values())[2]) # ELEMENT 03
31 ele_force_04 = np.array(list(ele_force.values())[3]) # ELEMENT 04
32 ele_force_05 = np.array(list(ele_force.values())[4]) # ELEMENT 05
33 ele_force_06 = np.array(list(ele_force.values())[5]) # ELEMENT 06
34 ele_force_07 = np.array(list(ele_force.values())[6]) # ELEMENT 07
35 ele_force_08 = np.array(list(ele_force.values())[7]) # ELEMENT 08
```

...ORS_04_COLUMNS_8_ANA\PUSHOVER_MDOF_10_FLOORS_04_COLUMNS

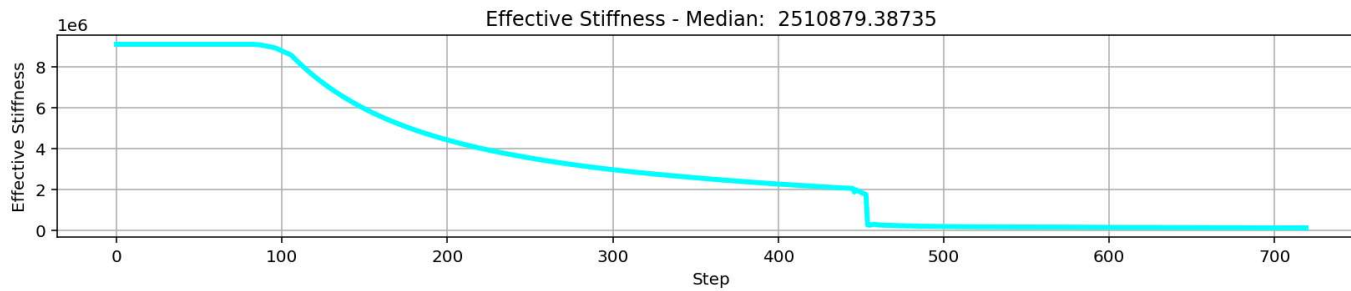




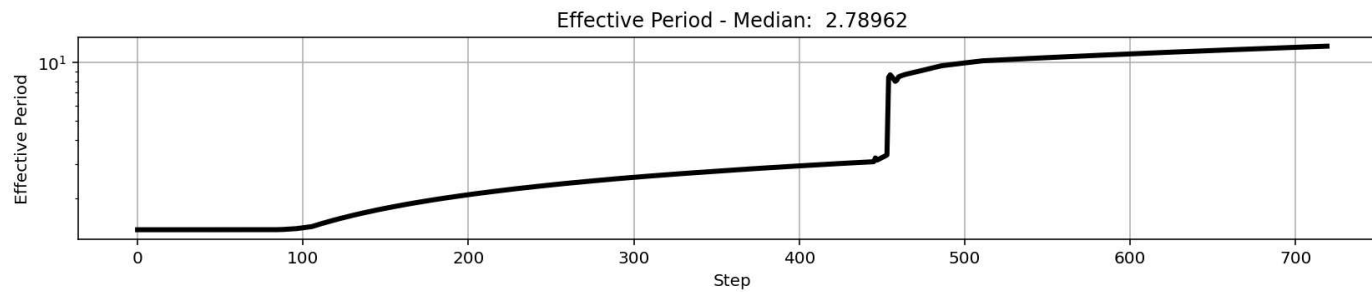
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

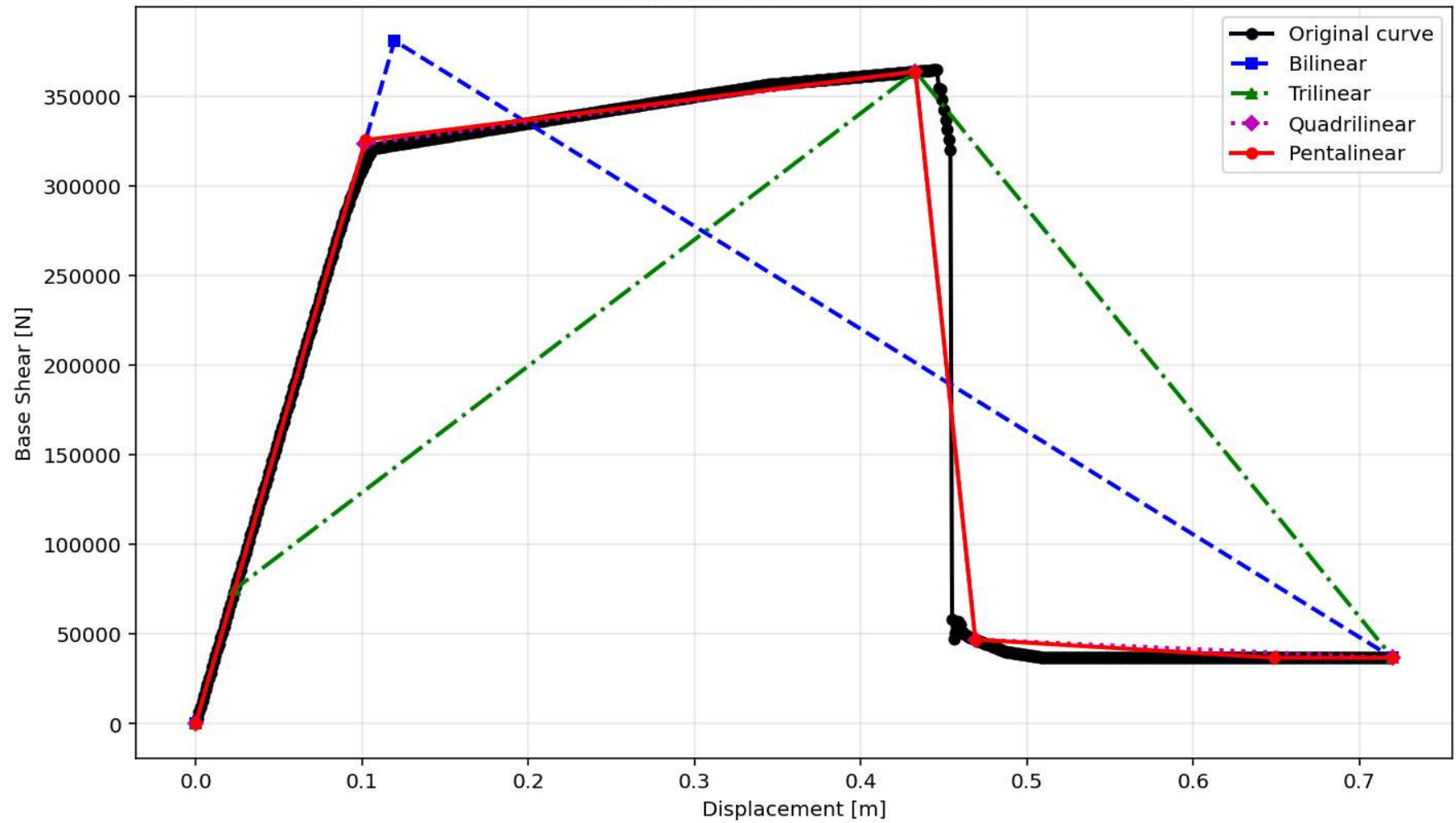


$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$



$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

Multilinear Fitting of Pushover Curve



BILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.19655567e-01 3.80722259e+05]

[7.20000000e-01 3.68200000e+04]]

+=====+

+-----+

Structure Elastic Stiffness : 3181818.18

Structure Plastic Stiffness : 51138.89

Structure Tangent Stiffness : [3181818.18 -572841.59]

Structure Ductility Ratio : 6.02

Structure Over Strength Factor: 0.10

+-----+

TRILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[2.35630358e-02 7.49732957e+04]

[4.33000000e-01 3.63556135e+05]

[7.20000000e-01 3.68200000e+04]]

+=====+

+-----+

Structure Elastic Stiffness : 3181818.18

Structure Plastic Stiffness : 51138.89

Structure Tangent Stiffness : [3181818.18 704828.5 -1138453.43]

Structure Ductility Ratio : 30.56

Structure Over Strength Factor: 0.49

+-----+

QUADRILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.01558313e-01 3.23140086e+05]

[4.33000000e-01 3.63556135e+05]

[4.69000000e-01 4.69601085e+04]

[7.20000000e-01 3.68200000e+04]]

+=====+

+-----+

Structure Elastic Stiffness : 3181818.18

Structure Plastic Stiffness : 51138.89

Structure Tangent Stiffness : [3181818.18 121940.15 -8794334.08 -40398.84]

Structure Ductility Ratio : 7.09

Structure Over Strength Factor: 0.11

+-----+

PENTILINEAR CURVE

+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.02268201e-01 3.25398820e+05]

[4.33000000e-01 3.63556135e+05]

[4.69000000e-01 4.69601085e+04]

[6.49000000e-01 3.68200000e+04]

[7.20000000e-01 3.68200000e+04]]

+=====+

+-----+

Structure Elastic Stiffness : 3181818.18

Structure Plastic Stiffness : 51138.89

Structure Tangent Stiffness : [3181818.18 115372.38 -8794334.08 -56333.94 0.]

Structure Ductility Ratio : 7.04

Structure Over Strength Factor: 0.11

+-----+